

Question	Answer	Marks
8(a)	region where there is a force	M1
	experienced by a current-carrying conductor/moving charge/(permanent) magnet	A1
8(b)(i)	single path, deflection in 'upward' direction	B1
	acceptable circular arc in whole field	B1
	no 'kinks' at start or end of curvature, and straight outside region of field	B1
8(b)(ii)	force (on particle) is normal to velocity/direction of motion/direction of speed	B1
8(c)	magnetic force provides/is the centripetal force	B1
	$Bqv = mv^2/r$ or $r = mv/Bq$	C1
	(if q is doubled), new speed = $2v$	A1

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9(a)	(induced) e.m.f. proportional/equal to rate	M1